

SKILLPULSE CLUSTER INFRASTRUCTURE TROUBLESHOOTING REPORT

Classification: Technical DevOps Portfolio Asset

Namespace: skillpulse

Audience: Technical Interviewers / Infrastructure Teams

1. Database Connection Loop

Symptom: Backend stuck at 0/1 Running waiting for DB.

Reason: Race condition and DNS/secret lookup issues.

Commands: `kubectl apply -f 20-backend.yaml -n skillpulse`; `kubectl delete pods -l app=backend -n skillpulse`

2. Manifest Validation Exception

Symptom: cannot convert int64 to string.

Reason: Env values required quoted strings.

Commands: Rewrite manifest via EOF and reapply.

3. Application Routing Deadlock

Symptom: Running but 0/1 READY.

Reason: Missing /health endpoint for readiness/liveness probes.

Commands: Patch deployment to remove probes and recreate backend pods.

Final Verification

```
kubectl get pods,svc -n skillpulse
backend 1/1 Running
frontend 1/1 Running
mysql 1/1 Running
backend svc :8080
frontend svc :8080
mysql svc :3306
```

Infrastructure State: 1/1 READY and Fully Operational.